HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed embedded C/C++ software for hardware-integrated edge intelligence platforms, focusing on signal processing and analytics workloads.
- Designed and implemented communication infrastructure for deploying machine learning models on embedded hardware, ensuring efficient data flow.
- Contributed to the design and optimization of Digital Signal Processing (DSP) algorithms, integrating hardware acceleration capabilities for enhanced performance.
- Owned components of solution development from technical requirements through implementation and validation, coordinating with cross-functional teams for project delivery.
- Participated in agile development cycles, including backlog definition and sprint planning, to ensure alignment with project objectives and timely execution.
- Explored and evaluated emerging technologies and methodologies relevant to embedded systems and edge analytics, contributing to team knowledge sharing.
- Containerized embedded software components using Docker for consistent deployment and integrated them with Jenkins for automated build and test pipelines.

SKILLS

Embedded Systems – Embedded C/C++, Embedded Linux, RTOS, Signal Processing, DSP, FFT, DWT, Filtering, Hardware Integration, Schematics, Drivers, SPI, I2C, DMA, System Architecture, System Integration, Performance Optimisation

Development Tools – Git, Build Systems, Cross-Compilation, Multi-Repository, Debugging, Problem-Solving, MATLAB/Simulink, Unit Testing, TDD, CI/CD, GitHub Actions, QEMU, Renode

Cloud & Platforms – AWS, EC2, S3, Lambda, CloudWatch

Programming Languages – C, C++, Python, Java

Methodologies – Agile, Scrum, DFSS, Lean, Secure Coding, Design Patterns, SOLID

Data & Analytics – Machine Learning, Data Science, Analytics Workloads

PROJECTS

Edge Intelligence Platform Development

- Designed and implemented embedded software for a next-generation edge intelligence platform, focusing on real-time data processing and analytics.
- Developed hardware-integrated solutions for deploying machine learning models on embedded devices, including signal processing algorithm optimization.
- Integrated software components with hardware schematics and drivers, ensuring reliable communication via protocols like SPI and I2C.

Automated Embedded Systems Validation Framework

- Built software-in-the-loop (SIL) and processor-in-the-loop (PIL) validation frameworks to test embedded software functionality and performance.
- Developed Python scripts to automate test execution and data collection within the validation frameworks, reducing manual testing effort by approximately 40%.
- Managed source code using Git and configured Jenkins pipelines for continuous integration and automated testing of embedded system builds.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)